

Harshita Bhasin

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PROFILE SUMMARY

Curious and driven IT undergraduate with hands-on experience in React.js, Flask, REST APIs, and AI/ML-based projects. Interested in research-oriented work involving practical problem-solving, experimentation, and building user-focused software solutions. A quick learner and collaborative teammate with strong attention to detail and a passion for learning emerging technologies.

TECHNICAL SKILLS

Frontend: HTML, CSS, JavaScript, React.js, Responsive Design

Backend: Flask, FastAPI, REST APIs

Libraries: PyTorch, TensorFlow, OpenCV, Librosa, NumPy

Developer Tools: Git, GitHub, Docker, Hugging Face Spaces

EXPERIENCE

Intern | Airports Authority of India (AAI, IT CHQ)

July 2025 – August 2025

AES-RSA Cryptographic Performance Analysis | GitHub Repository

- Built a **Flask-based HTTPS testbed** to compare **AES vs RSA** encryption-decryption workflows in a controlled localhost environment.
- Benchmarked **AES across 6 sessions**, with **6/6 runs below the 70,000 ms benchmark** and timings ranging from **33,000–59,000 ms**.
- Result:** Evaluated **12 total sessions**; **AES** completed **all runs below benchmark**, while **RSA** showed **2 benchmark breaches**, reinforcing AESs efficiency for bulk encryption.

PROJECTS

Shelf | React, Vite, JavaScript, CSS, REST API, Tesseract.js

GitHub | Live Demo

- Developed a **React + Vite** book discovery app with **2 search workflows**: **text-based search** using **Google Books API** and **Tesseract.js OCR-assisted image search**.
- Implemented **OCR-based uploads** to extract **text** from book pages/screenshots and enhance discovery through a **responsive UI**.

Deepfake Detection Model | Python, ResNet18, FastAPI, Docker, JavaScript

GitHub | Live Demo

- Built a deepfake detection web application with a **FastAPI inference backend**, enabling users to upload image, audio, and video files and view prediction results through a responsive frontend interface.
- Configured **PyTorch-based inference pipelines** using a **ResNet18 image classifier**, **MFCC-based audio classifier**, and **CNN+LSTM lip-sync model**.
- Evaluated the audio model on an **800-sample validation setup**, achieving **85.6% validation accuracy**, and deployed the Dockerized backend with a Vercel-hosted frontend.

EDUCATION

Bharati Vidyapeeth's College of Engineering, New Delhi

B.Tech in Information Technology, Specialization in Data Science

2023 - 2027

CGPA: 9.23

ACHIEVEMENTS

- Secured **1st Position** in Pixel Planet at BVEST 2026
- Secured **1st Position** in MicroQuest at BVEST 2025

POSITION OF RESPONSIBILITY

Treasurer, IEEE BVCOE: Managed budgeting, financial coordination, and event-related expenditure tracking for student branch activities